

Introduction to Software Engineering (COSC 336)

LAB PHASE 1

Initial Plan and Requirement Gathering Document

Intelligent, AI-powered Circular Campus Resource Exchange And Asset Life Cycle Management System

LAB INSTRUCTORS:

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Project Overview

- Lab-based Running Project
- **Project Title:** Intelligent, AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System
- **Objective:** Apply the complete software development life cycle while comparing classic and AI-enhanced engineering approaches.
- **Project Focus:** Build an internal platform that identifies, classifies, matches, transfers, maintains, and tracks campus resources throughout their life cycle.



Project Overview

- Universities may purchase new furniture, computers, laboratory equipment, books, and office materials while usable items remain idle elsewhere.
- Traditional asset management systems mainly record:
 - Ownership and location
 - Purchase information
 - Disposal status

However, they often **do not** support:

- Discovering and classifying reusable assets
- Matching assets with departmental needs
- Coordinating transfers and maintenance
- Tracking assets throughout their life cycle
- Predicting future demand
- Measuring financial and environmental impact

An integrated system is needed to connect available campus resources with departmental demand and manage them throughout their life cycle.

Phase	Submission deadline	Duration	Deliverable	Weight
Phase 1	In the labs of the week of Sep 14 th , 2026	1 Week	Initial Plan and Requirement Gathering Document	10%
Phase 2	In the labs of the week of Sept 21 st , 2026	1 Week	Feasibility Document	10%
Phase 3	In the labs of the week of Oct 5 th , 2026	2 Weeks	Requirements Document	10%
Phases 4 & 5	In the labs of the week of Oct 26 th , 2026	3 Weeks	Design Document (Architecture and Detailed Design)	20%
Phase 6	In the labs of the week of Nov 16 th , 2026	3 Weeks	Draft Implementation with Major Features	20%
Phase 7	In the labs of the week of Nov 23 rd , 2026	1 Week	Details of Test Cases	10%
Phase 8	In the labs of the week of Nov 23 rd , 2026		Final Project, Presentation and Demonstration	20%

Project Phases, Deliverables and Weights

Classic Functionalities:

1. **Asset Registration and Publication:** Register and maintain campus asset records, and publish surplus, underutilized, or unwanted resources.
2. **Resource Request and Reservation:** Enable departments to submit resource requests and to search, filter, reserve, and request available resources.
3. **Approval and Transfer Management:** Manage approvals and asset transfers, including changes in ownership, custody, and location.
4. **Inspection and Asset Life-Cycle Tracking:** Record inspections, maintenance, repairs, donations, recycling, and disposal actions.
5. **User and Access Management:** Manage user accounts with clearly defined roles and permissions.

AI-powered functionalities

1. **AI-Based Resource Matching:** Match asset listings with departmental requests using semantic similarity, technical compatibility, condition, quantity, location, and urgency.
2. **AI-Assisted Asset Classification:** Recommend appropriate categories, tags, and standardized descriptions for newly registered resources.
3. **Sustainability Optimization:** Recommend whether an asset should be reused, transferred, repaired, refurbished, donated, recycled, or retired.
4. **LLM-Powered Assistant:** Enable users to search for resources, submit requests, ask asset-related questions, and navigate system services using natural language.
5. **Generative AI Reporting:** Summarize resource-exchange activities, maintenance trends, avoided purchases, financial savings, waste diversion, and estimated reductions in carbon emissions.

Phase 1: Planning and Requirement Gathering

- Define the project **problem, scope, objectives, assumptions, and expected benefits.**
- Identify **stakeholders** such as department representatives, asset custodians, requesters, administrators, etc.
- Map stakeholders to specific system features, data needs, permissions, responsibilities, and approval decisions.
- Study the current process for identifying surplus assets, requesting resources, approving transfers, recording maintenance, and managing donation, recycling, or disposal.

Phase 1: Planning and Requirement Gathering

- Define **measurable project success criteria**, including:
 - functional performance
 - user acceptance
 - match usefulness
 - avoided purchases
 - reuse, repair, and sustainability outcomes.
- Develop a preliminary **project plan** with:
 - timelines
 - milestones
 - task assignments
 - dependencies, risks, and resource allocations.

Phase 1: What to Submit

Title Page

- Title of the project
- Date of the document
- Prepared by (author's name)
- Prepared for (client's or company's name)

Introduction

- Project concept
- Document purpose

Project Overview

- High-level description of the proposed system or software
- Background and context of the project
- Business goals and objectives
- Project constraints (budget, timeline, resources)
- **Scope of the project**

Stakeholder Analysis

- Identification of all stakeholders (users, clients, team members, etc.)
- Stakeholder roles and responsibilities
- Stakeholder needs and expectations

Phase 1: What to Submit

Requirements Gathering

- Methodology for gathering requirements (interviews, surveys, workshops)
- Summary of gathered requirements
- Prioritization of requirements

Initial Requirements Specification

- Detailed description of functional/operational requirements

Initial Project Plan Outline

- Preliminary project timeline
- Major milestones and deliverables
- Proposed project methodology (Agile, Waterfall, etc.)
- Resource allocation plan (team structure, roles)

Important: Read the complete project description carefully. The slides provide only a summary. All requirements, functionalities, constraints, and deliverables specified in the project description, including details not repeated in the slides, are part of the project and may be assessed.

Important Guidelines

- This project must be developed using Git and hosted on **GitHub**.
- Each team is required to create a GitHub repository for the project and add the instructor's and the lab engineer's GitHub usernames as collaborators **by September 15**.
- Development progress must be tracked through regular, meaningful commits throughout the project period.
- **Submitting the entire final project as a single commit does not satisfy this requirement and will result in a grade penalty.**
- GitHub commit history will be reviewed to verify the timing, consistency, and individual authorship of project work. It will account for **10%** of each student's phase grade.
- Repositories with insufficient or irregular commit activity, or with no clear individual authorship in team projects, will be flagged for review.

Thank you!